SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence   COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 30 Minutes

QUESTIONS: 30   **Total Marks:30**

Annexure A

MCQ

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Q1. Which of the following best describes a Logical Agent in AI?

a) An agent that reacts randomly to the environment c) An agent that only follows pre-defined scripts
b) An agent that uses a knowledge base and logical inference to decide actions d) An agent that learns from rewards

Q2. In Propositional Logic, 'P → Q' is logically equivalent to:

a) $P \land \neg Q$ c) $\neg Q \rightarrow \neg P$ only
b) $\neg P \lor Q$ d) Both b and c

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:
a) A disjunction of conjunctions c) A conjunction of literals only
b) A conjunction of disjunctions (clauses) d) A sequence of implications

Q4. In FOL, the sentence $\forall x$: Student(x) → Passes(x) means:
a) Some students pass c) If something passes, it is a student
b) Every entity that is a student passes d) There exists at least one student who passes

Q5. Unification in FOL is the process of finding substitutions that:

a) Convert clauses to CNF
b) b) Make two logical expressions syntactically identical

c) Prove a goal using backward chaining
d) Derive facts using Modus Ponens

6.What is the core function of a logical agent in AI?

a) Reacting only to current percepts

b) Using logic and knowledge representation to reason and make decisions

c) Performing physical actions without internal representation

d) Relying solely on sensors for navigation

7.Which component of a logical agent is responsible for storing facts and rules about the world?

a) Inference Engine

b) Perception Unit

c) Knowledge Base (KB)

d) Action Selector

8.What does a knowledge-based agent do immediately after obtaining a percept from the environment?

a) Selects an action

b) Executes the action

c) Updates the Knowledge Base with the new percept

d) Reasons to derive new information

9.Which operation is used to add new information to a Knowledge Base?
a) ASK b) TELL c) PERCEIVE d) UPDATE

10.What is a proposition in formal logic?
a) A command or request
b) A declarative statement that is either true or false
c) A question regarding the environment
d) A letter representing a variable

11.Which of the following is NOT a proposition?
a) "The sun rises in the east."
b) "5 + 3 = 10."
c) "Close the door."
d) "Chennai is in India."

12.Which logical operator is represented by the symbol ¬?
a) AND
b) OR
c) NOT
d) Implication

13.The AND (∧) operator is true only when:
a) At least one proposition is true
b) Both propositions are true
c) Both propositions have different truth values
d) Neither proposition is true

14.A proposition that is always true, regardless of the truth values of its variables, is called a:
a) Contradiction b) Tautology c) Contingency d) Equivalence

15.In an implication (P → Q), the statement is false only when:
a) P is false and Q is true
b) P is true and Q is false
c) Both P and Q are false
d) Both P and Q are true

16.What is the number of rows in a truth table for a logical expression with 3 propositions?
a) 4 b) 6 c) 8 d) 9 An

17.Which of these represents one of De Morgan's Laws?
a) $\neg(P \land Q) \equiv \neg P \lor \neg Q$
b) $\neg(P \land Q) \equiv \neg P \land \neg Q$

c) $\neg(\neg P) \equiv P$

d) $P \lor \neg P \equiv$ True

18.A proposition that is sometimes true and sometimes false is known as a:

a) Tautology b) Contradiction c) Contingency d) Implication

19.Which logical connective is true if and only if both propositions have the same truth value?

a) Implication (→) b) Biconditional (↔) c) OR (∨) d) NOT (¬)

20.According to operator precedence, which operator should be evaluated first?

a) AND (∧) b) OR (∨) c) NOT (¬) d) Implication (→)

21.What is the process of deriving a conclusion from given premises called?

a) Perception b) Representation c) Inference d) Predication

22.Which inference rule follows the form: If P, then Q; P; Therefore, Q?

a) Modus Tollens b) Modus Ponens c) Hypothetical Syllogism d) Resolution

23.Which valid reasoning pattern is represented by: $P \to Q$; $\neg Q$; Therefore, $\neg P$?

a) Modus Ponens b) Disjunctive Syllogism c) Modus Tollens d) Addition

24.What can be concluded from the premises $(P \to Q)$ and $(Q \to R)$ using Hypothetical Syllogism?
a) $P \land R$ b) $P \lor R$ c) $P \to R$ d) $\neg P \to \neg R$

25.The rule that states if "P or Q" is true and "Not P" is true, then "Q" must be true is called:

a) Simplification b) Conjunction c) Disjunctive Syllogism d) Resolution

26.Which rule allows you to conclude "P" from the premise "P and Q"?

a) Addition b) Simplification c) Conjunction d) Modus Ponens A

27.The inference rule Resolution follows which symbolic form?

a) $P \therefore P \lor Q$ b) $P, Q \therefore P \land Q$ c) $P \lor Q, \neg P \lor R \therefore Q \lor R$ d) $P \to Q, Q \to R \therefore P \to R$

28.Which of the following is an example of the "Addition" rule?

a) P and Q, therefore P

b) P, therefore P or Q

c) P, Q, therefore P and Q

d) P or Q, not P, therefore Q

29."If it rains, the ground is wet. The ground is wet. Therefore, it rained." This is an example of which invalid reasoning pattern?

a) Denying the Antecedent b) Affirming the Consequent c) Modus Ponens d) Double Negation

30.What is the name of the invalid pattern: $P \rightarrow Q$; $\neg P$; Therefore, $\neg Q$?
a) Affirming the Consequent b) Denying the Antecedent c) Modus Tollens d) Hypothetical Syllogism

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

12/08/26

CO3-AT2-MIQ

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CSA0703 - A1

1) b
2) d
3) b
4) b
5) ~~d~~ b
6) b
7) c
8) c
9) b
10) b
11) b.
12) c
13) b
14) b
15) b
16) c
17) a
18) c
19) b
20) c
21) c
22) b
23) c
24) c
25) c
26) c
27) c
28) b
29) b
30) b

8

28/30